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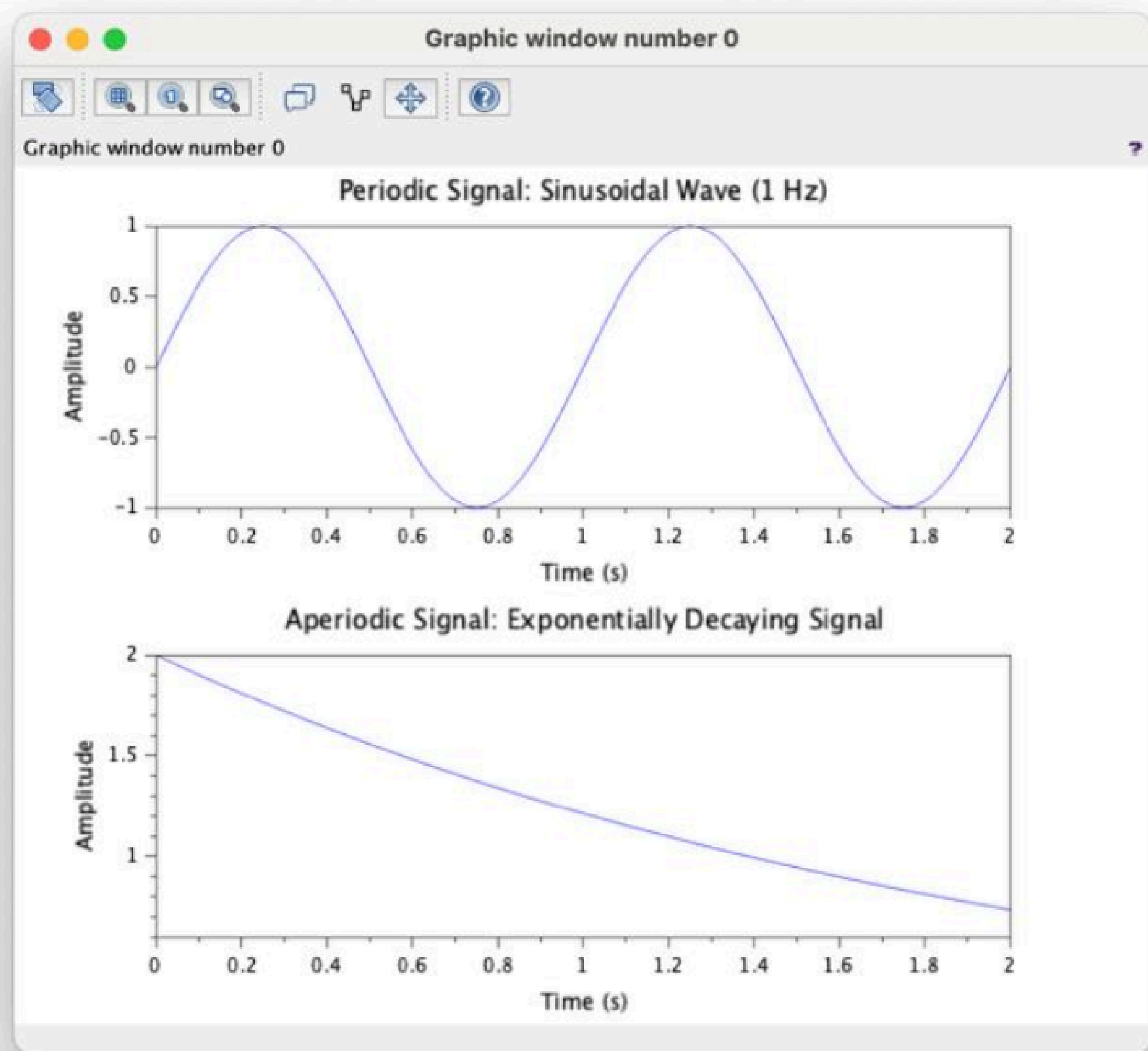
// Test Case: Generation of Periodic and Aperiodic Signals
// Parameters for the periodic signal (Sinusoidal)
f_periodic = 1; // Frequency in Hz
A_periodic = 1; // Amplitude
t_periodic = 0:0.01:2; // Time vector for 0 to 2 seconds
// Generate the periodic signal
x_periodic = A_periodic * sin(2 * %pi * f_periodic * t_periodic);
// Plot the periodic signal
clf; // Clear the current figure
subplot(2,1,1); // Subplot 1 for the periodic signal
plot(t_periodic, x_periodic);
xlabel('Time (s)');
ylabel('Amplitude');
title('Periodic Signal: Sinusoidal Wave (1 Hz)');
// Parameters for the aperiodic signal (Exponentially Decaying)
a_aperiodic = 0.5; // Base of the exponential decay
A_aperiodic = 2; // Amplitude
t_aperiodic = 0:0.01:2; // Time vector for 0 to 2 seconds
// Generate the aperiodic signal
x_aperiodic = A_aperiodic * exp(-a_aperiodic * t_aperiodic);
// Plot the aperiodic signal
subplot(2,1,2); // Subplot 2 for the aperiodic signal
plot(t_aperiodic, x_aperiodic);
xlabel('Time (s)');
ylabel('Amplitude');
title('Aperiodic Signal: Exponentially Decaying Signal');
```



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t = -1:0.001:1;
x = exp(-abs(t)).*sin(4*%pi*t) + 0.3;
// Reverse signal
x_rev = interp1(t, x, -t, 'linear');
// Even and odd parts
xe = 0.5 * (x + x_rev);
xo = 0.5 * (x - x_rev);
clf;
plot(t, x, 'k');
plot(t, xe, 'b--');
plot(t, xo, 'r:');
legend("x(t)", "Even Part", "Odd Part");
xtitle("Even-Odd Decomposition", "t", "Amplitude");
```